

CS 219

Lecture 31

the file abstraction

- persistent storage  $\nabla$  ephemeral storage
- egs: file, key-value, RDBMS, NoSQL DBS, ...

OS abstraction for persistent storage

features: naming, access control, timestamps, size...

: operations  $\sim$  open, read, write, truncate

is a linear <sup>ordered</sup> sequence of bytes.

organization of files

hierarchical namespace.

(\*) the File System is the OS component for the file abstraction.

msdos, NTFS,

ext2, ext4, ~~re~~ btrfs, - - -

## # requirements/features of a FS.

- provide the file abstraction
- management actions
  - ↳ reliability, efficiency, capacity/utilization...
- uniformity of access interface.

### - HDD

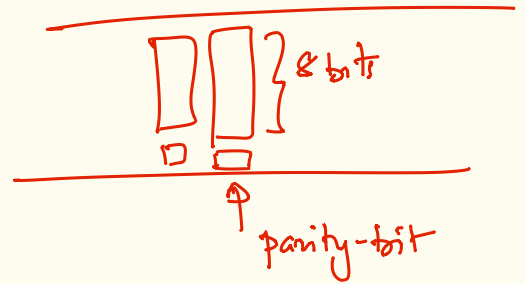


- head, cylinder, track, sector,
- moving parts

### SSD

- NAND/NOR gates
- persistency
- no moving parts

### TAPE



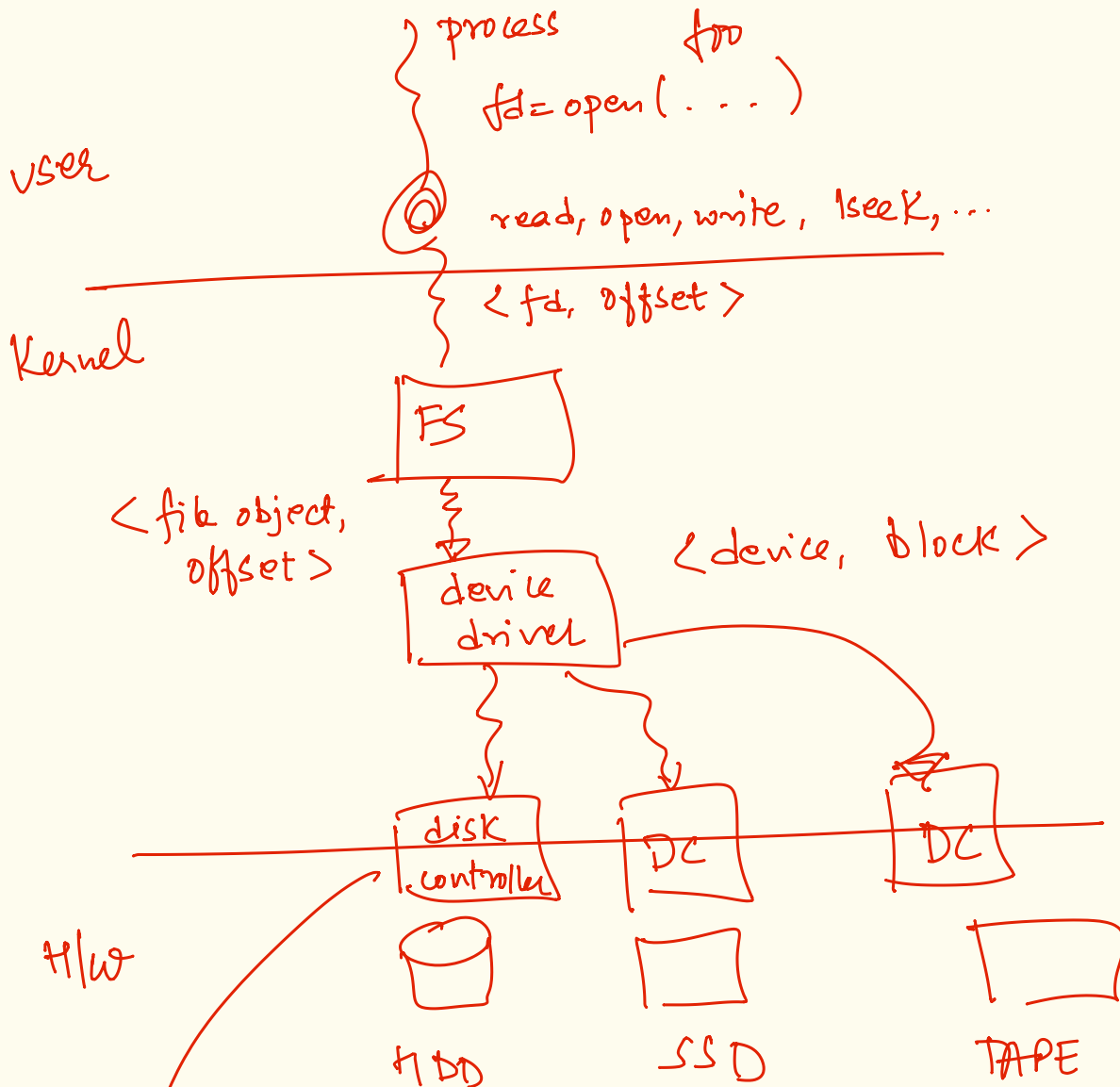
② the process-view of files or file systems  
listing of files & directories.  
~ process use/engage with a file;  
↳ a file descriptor ~ an abstraction to "refer" to a file;

fd = open(filename, mode, ...)

read(fd, buf, size);

write(fd, buf, size);

} all fds stored in fdtable associated w/ the process.  
in the PCB.



exposes a block abstraction

512 to 4096 bytes

Storage system is a linear ordered set of blocks.

disk

file foo



offset 4,

file, 4

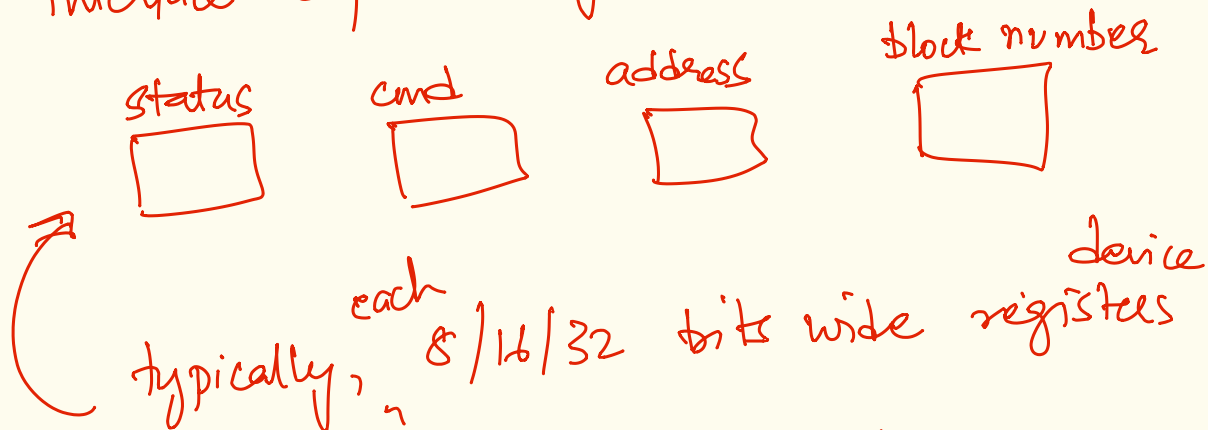
fd, 4

file object, 4

device, block 2023

## #1 Simple device driver

interface exposed by the Controller —



— mapped to memory (mmio)

— port-mapped IO — in/out port numbers & sequencing

driver:

```
while (status == BUSY) { }
```

```
CMD ← R
```

```
addr ← Kva of a buffer
```

destination  
memory  
address.

```
block ← block no. to read
```

```
STATUS ← BUSY
```

custom command to the controller.